Splunk SPL Search Queries

Authentication Brute-Force Investigation — Lab 1

Search 1: Aggregate Failed Logons to Find the Attacker IP
Splunk SPL Command :

```
index="soc-investigation" source="*auth_bruteforce_lab1.csv*" EventID="4625"
| dedup _time, ComputerName, Account_Name, Source_Network_Address
| stats count as Failed_Attempts values(Failure_Reason) as Reason by Source_Network_Address, Acc
ount_Name, ComputerName
| sort - Failed_Attempts
```

Explanation

• Searches authentication logs for Windows Security Event ID 4625 (Failed Logon).

• Removes duplicate rows in case the CSV file was uploaded multiple times.

• Groups failed login attempts by source IP, target username, and server name (SRV-WEB-01), counting total failures and listing the failure reason (Bad Password or Unknown User).

• Sorts results from highest to lowest failed-attempt count, placing the attacker IP 198.51.100.44 targeting Administrator above normal 1–2 count password typos.

Search 2: Pivot on Attacker IP to Confirm Breach

Splunk SPL Command :

```
index="soc-investigation" source="*auth_bruteforce_lab1.csv*" Source_Network_Address="198.51.100
.44"
| dedup _time, EventID, Account_Name
| eval Event_Description=if(EventID="4625", "4625 - Failed Logon", "4624 - SUCCESSFUL BREACH")
| sort _time
| tail 10
| table _time, ComputerName, Account_NameSource_Network_Address, Logon_Type, Failure_Reason,
 Event_Description,
```

Explanation

• Filters every authentication event originating from the attacker IP 198.51.100.44, including both 4625 failures and 4624 successes.

• Creates an Event_Description label: “4625 - Failed Logon” for failed events and “4624 - SUCCESSFUL BREACH” for successful events.

• Sorts events chronologically from oldest to newest.

• Keeps only the last 10 events so the transition from failed attempts (10:01:10–10:01:42) to the successful breach (10:01:45) fits cleanly in one view.

• Displays Logon_Type=10, indicating RemoteInteractive / RDP access.

Search 3: Brute-Force Spike Over Time

Splunk SPL Command :

```
index="soc-investigation" source="*auth_bruteforce_lab1.csv*" EventID IN ("4624", "4625")
| dedup _time, EventID, Account_Name, Source_Network_Address
| eval Result=if(EventID="4625", "Failed (4625)", "Success (4624)")
| timechart span=1m count by Result
```

Explanation

• Includes both successful and failed logon events.

• Converts numeric EventIDs into readable chart legend labels: “Failed (4625)” and “Success (4624)

Groups events into one-minute buckets to visualize the brute-force burst between 10:00 and 10:02 on a Column Chart.